

Top-k Ranking Preservation under Bounded Score Perturbations

Research Architecture Runtime
`operator/top-k`

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Abstract

Top- k selection is certified via pairwise ranking preservation when all gaps exceed 2ε . Lean:
`LEAN_FULL`.

1 Theorems

Theorem 1 (Preservation). *All pairwise gaps $> 2\varepsilon$ imply $s_i < s_j$ iff $(s+\delta)_i < (s+\delta)_j$ for $\|\delta\|_\infty \leq \varepsilon$, hence the top- k index set is invariant.*

Theorem 2 (Sharpness). *A pairwise gap $\leq 2\varepsilon$ admits a collision adversary.*

2 Lean / certificate

`Research.Operators.TopK.Preservation` reducing to `OrderStat.Ranking`.

3 Limitations

Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).